PRACTICAL 12

Aim :

Set up a GitHub Actions CI/CD pipeline for the Task Management application. The pipeline must install dependencies, run automated backend tests, build the frontend, and report a clear pass or fail result on pushes and pull requests targeting the main branch.

Code :

.github/workflows/ci.yml :-

```
name: CI Pipeline

on:
  push:
    branches: [main]
  pull_request:
    branches: [main]

jobs:
  backend-test:
    runs-on: ubuntu-latest
    defaults:
      run:
        working-directory: backend
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-node@v4
        with:
          node-version: 20
          cache: npm
          cache-dependency-path: backend/package-lock.json
      - run: npm ci
```

```
    - run: npm test

  frontend-build:

    runs-on: ubuntu-latest

    defaults:

      run:

        working-directory: frontend

    steps:

      - uses: actions/checkout@v4

      - uses: actions/setup-node@v4

        with:

          node-version: 20

          cache: npm

          cache-dependency-path: frontend/package-lock.json

      - run: npm ci

      - run: npm run build
```

Code :

backend/package.json test script :-

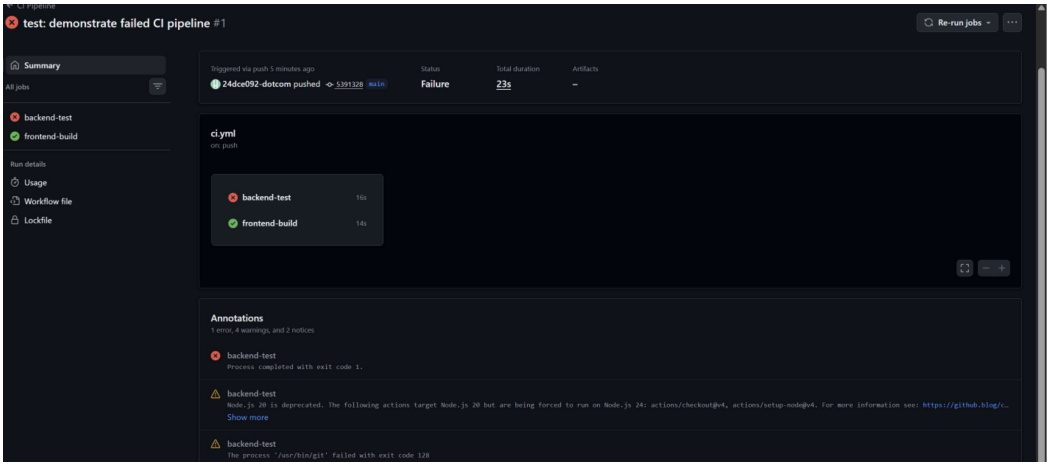
```
"scripts": {
  "test": "jest --runInBand"
}
```

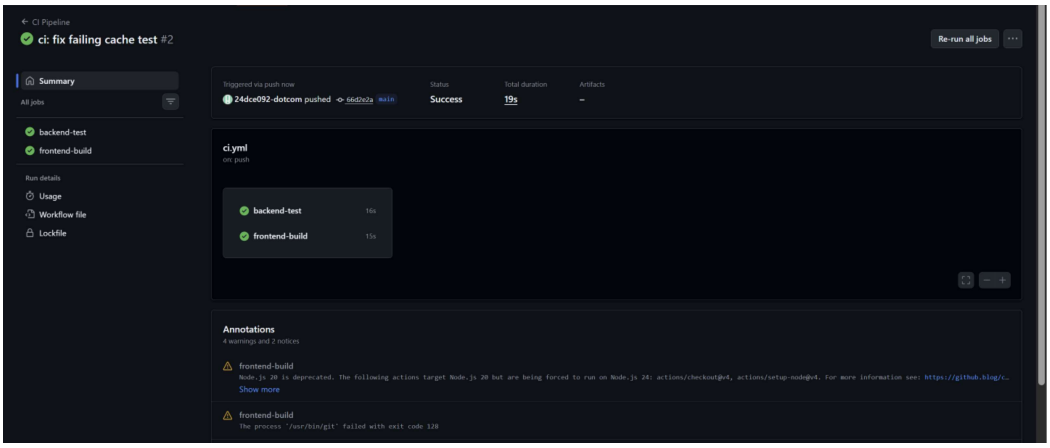
backend/test/cache.test.js :-

```
const cache = require('../cache');
describe('task cache', () => {
  beforeEach(() => {
    cache.invalidateTaskCaches();
  });
  test('stores and returns a cached value', () => {
    const tasks = [{ _id: 'task-1', title: 'CI task' }];
    cache.set('tasks:all', tasks);
    expect(cache.get('tasks:all')).toEqual(tasks);
  });
```

```
  test('invalidates collection and single-task keys', () => {
    cache.set('tasks:all', [{ _id: 'task-1' }]);
    cache.set('tasks:task-1', { _id: 'task-1' });
    cache.invalidateTaskCaches('task-1');
    expect(cache.get('tasks:all')).toBeUndefined();
    expect(cache.get('tasks:task-1')).toBeUndefined();
  });
});
```

Output :





Result :

The GitHub Actions CI pipeline was successfully configured. Backend tests and the frontend production build run automatically on pushes and pull requests targeting main. The workflow reports clear pass or fail status and catches regressions before integration.